
THE FLOW LANDSCAPE — SYSTEMATIC & DEALER FLOWS IN THE UNDERLYING

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Abstract

Prices in a liquid index do not move only because opinions about value change; they move because, at any moment, some participants *must* trade regardless of price. This paper is the integrating map of the library: it collects the large systematic and dealer flows in the equity-index underlying, puts them into a single comparable unit, and asks which one dominates in which regime. We build a taxonomy along one organizing axis — how free a participant is to walk away from a bad price — and argue, following the inelastic-markets literature, why price-insensitive demand makes an index far more moveable than efficient-market intuition suggests. Price impact, and specifically the square-root law, is developed as the common currency that lets a dealer gamma hedge be compared against a volatility-target rebalance. We then treat four flow families in turn: dealer re-hedging (an embedded recap of the mechanics from the dealer-flows-and-GEX paper — gamma, charm, and vanna as a flow), volatility-target and risk-parity deleveraging, CTA trend-following, and passive and calendar flows. Each is reconstructed on free data with a single pre-committed rule, and the numbers are reported honestly whether or not they flatter the story — the turn-of-month test, in particular, decays out-of-sample. We close with a flow calendar and regime map that combines these flows with the volatility-factor gate from the volatility-surface paper, a candid catalogue of identification limits (flows are estimated, overlapping, endogenous, and non-stationary), and a hand-off in which each flow becomes one pre-registered, falsifiable hypothesis. Every figure is reproducible from the accompanying code on freely available data.

Keywords — Order flow, price impact, square-root law, inelastic markets, dealer hedging, gamma exposure, volatility targeting, risk parity, CTA, trend following, time-series momentum, turn-of-month, index rebalancing, regime map, market microstructure, identification

1 Introduction

1.1 Motivation and thesis

The reflex taught by the efficient-market tradition is that a price is a discounted stream of fundamentals, and that any predictable pattern in *flow* — who is buying, who is selling, and when — is competed away by arbitrageurs until it leaves no footprint. Under that view the identity of the marginal buyer is irrelevant: demand curves for a broad index are nearly flat, because close substitutes exist and informed capital is abundant, so a large purchase moves price only to the extent it conveys news.

A growing and, by now, hard-to-dismiss body of evidence says the opposite is closer to the truth. Demand curves for the aggregate equity market are *steep*, not flat; a dollar of price-insensitive buying moves aggregate market value by several dollars, not by nothing (Gabaix & Koijen 2021). If that is right, then the identity and the compulsion of the marginal trader matter enormously, and a serious account of *how the underlying moves* cannot stop at "news arrived." It has to name the participants whose trading is, at least partly, forced — mechanically tied to price, to realized volatility, to the passage of time, or to the calendar — and it has to size those forced trades in a unit that lets them be compared.

That is the thesis of this paper, and it is deliberately modest about what it can prove. *The large moves in an index are, to a first approximation, a competition between flows that are not free to walk away from a bad price, and understanding the underlying means mapping those flows, sizing them in a common unit, and knowing which one dominates in which regime.* Dealer hedging — the subject of the dealer-flows-and-GEX paper — is one such flow, and an important one; but it lives next to volatility-target funds that deleverage into weakness, trend-followers that flip at reversals, and passive and calendar programs that trade on dates you can circle in advance. The value of a *map* is not a new prediction. It is comparability (a gamma hedge and a risk-parity rebalance measured on the same scale), timing (which flow is loudest in which regime), and — most of all — honesty about how little of this can be cleanly identified from price data alone.

1.2 Scope, prerequisites, and intended reader

This paper sits in the Trading Library's *Market Mechanics* track. It is the integrating synthesis of the track: where the how-markets-move paper explains why a price moves at all, this paper assembles the flows that push it. It draws on three other documents and states where. It assumes the dealer-hedging mechanics developed in depth in the dealer-flows-and-GEX paper, and treats that material here only in embedded recap. It uses the orthogonal volatility-factor set built in the volatility-surface paper as the regime gate in Section 8. And it takes its evidence discipline — pre-registration, the out-of-sample split, and the direction check — from the backtesting paper, handing the resulting hypotheses forward to *Overfitting Control & Forecast Calibration*.

The paper assumes comfort with returns and volatility, a little probability, and the ability to read a short Python snippet; it assumes no stochastic calculus. After reading it, a reader can:

- place any market participant on a single taxonomy by asking two questions — *what pulls the trigger* (information, price, volatility, or the clock) and *how free is the trader to walk away from a bad price*;

- translate a dollar flow into an expected price move using the square-root impact law, and use impact as a common unit to compare a dealer hedge against a CTA rebalance;
- state the dealer re-hedging demand as a signed, sized, timed flow, and say when it stabilizes the tape and when it destabilizes it;
- reconstruct a volatility-target deleveraging path and a CTA positioning proxy from free daily prices, with one pre-committed rule and no tuning;
- run a pre-committed turn-of-month test with confidence intervals and an out-of-sample split, and report what it shows even when that is a near-null;
- assemble a flow calendar and regime map that says which flow dominates in trend versus chop and in calm versus stress, gated by the volatility-surface paper's volatility factors;
- and recite the identification limits — estimation, overlap, endogeneity, non-stationarity — that keep all of this at the level of a *disciplined prior*, not a validated signal.

1.3 Data and reproducibility

A discipline carried from the backtesting paper runs through the whole paper: the *direction check*. Before any figure is allowed to support a claim, the *sign* of the data is checked against the sign of the claim, because a result of the right magnitude and the wrong sign is a falsification wearing a disguise, and absorbing one silently has cost this author dearly before.

All numerical examples use freely available data — daily S&P 500 and Treasury-ETF closes from the public Yahoo Finance chart API, cached locally so the figures rebuild offline — and are reproducible from the accompanying code in *figures/*. Each figure uses exactly *one* rule, fixed in advance; no threshold, window, or asset is searched for the reading that looks best, because that search is the very data-snooping the paper warns against. The private, licensed, and vendor data that inform the author's own trading — intraday gamma-exposure snapshots and the like — appear here only as *mechanism*, never as published values: no proprietary feed is redistributed, and no licensed number is printed. *This is a research and educational document, not investment advice.*

1.4 Organization of this paper

The paper is complete; every section below is written in full. Section 2 builds the taxonomy of flows and the organizing axis, and states the inelastic-markets argument for why forced demand moves an index so much. Section 3 develops price impact — temporary versus permanent, and the square-root law — as the common currency in which every later flow is measured. Section 4 is the embedded recap of dealer hedging as a flow (gamma, charm, vanna), cross-referencing the dealer-flows-and-GEX paper for the estimation mechanics and asking here only how it stacks against everything else. Sections 5, 6, and 7 treat the three other large flow families — volatility-target and risk-parity deleveraging, CTA trend-following, and passive and calendar flows — each reconstructed on free data with one pre-committed rule and reported honestly. Section 8 assembles the flow calendar and regime map, combines it with the volatility-factor gate from the volatility-surface paper, and lays out the identification limits without softening them. Section 9 turns each flow into a pre-registered, falsifiable hypothesis and describes the reproducible free-data pipeline, handing off to the backtesting paper and the overfitting-control paper. The paper closes with annotated references.

2 A taxonomy of flows

To compare flows we first need to sort them, and the sort has to be built on the property that actually governs how a flow moves price. That property is not *who* the trader is or *why* they claim to be trading; it is *how free they are to walk away from a bad price*. A hedge fund with a valuation view will simply not buy if the offer is unattractive — its demand is *price-elastic*, and it supplies patience to the market. A volatility-target fund whose risk model has just registered a spike must cut exposure *today*, at whatever price clears — its demand is *price-inelastic*, and it consumes liquidity regardless of level. The distance between those two behaviours is the whole subject of this paper.

2.1 Two questions, one axis

Any participant can be located by asking two questions. The first — *what pulls the trigger?* — sorts flows into four families by the kind of information that forces a trade:

- **Information and valuation.** Discretionary fundamental buyers and sellers, and quantitative value or factor programs, trade when their estimate of worth diverges from price. The trigger is a *view*.
- **Price moves.** Market makers and high-frequency inventory managers, and trend-followers, trade because the *price itself* moved: the former to mean-revert an inventory imbalance, the latter to chase an established direction.
- **Volatility moves.** Volatility-target and risk-parity funds, and the vanna component of dealer hedging, trade because *realized or implied volatility* changed, independently of direction.
- **Clock and calendar.** Buyback programs, pension and month-end rebalancing, index reconstitutions, passive creations and redemptions, and the charm component of dealer hedging trade because *the date arrived* — the trigger is deterministic in time.

The second question — *how free is the trader to walk away?* — places each family on a horizontal axis from *price-elastic* (far left: trades only when the price is attractive, and supplies liquidity) to *price-inelastic* (far right: must trade regardless of price, and demands liquidity). This axis, not the trigger, is what determines a flow's capacity to move price against the crowd. Figure 1 lays the families out on the grid.

A taxonomy of flows: what pulls the trigger (rows) x who must trade (axis)



Figure 1. A taxonomy of flows. Rows answer what pulls the trigger — information and valuation, price moves, volatility moves, clock and calendar — and the horizontal position answers the organizing question, how free the trader is to walk away from a bad price. Far left is price-elastic and liquidity-supplying (discretionary value buyers); far right is price-inelastic and liquidity-demanding (dealer gamma hedging on a short book, target-vol deleveraging, dated index rebalances). Positions on the axis are qualitative orderings, not measurements. Reproduce with figures/fig_flow_taxonomy.py.

Two features of the picture matter. First, *the same trigger can produce either elastic or inelastic behaviour*: a market maker responding to a price move is mean-reverting and elastic (they will fade a move if paid to), whereas a short-gamma dealer responding to the same price move is pro-cyclical and inelastic (they *must* buy the rally to stay hedged). The mechanism, not the label on the desk, decides the sign. Second, *discretionary and mechanical are not the same axis as elastic and inelastic*, though they correlate: a rule-bound CTA still exercises some discretion in execution, and a "discretionary" macro fund forced to meet a redemption is, for that trade, perfectly inelastic. The axis that governs impact is elasticity, and it is the one we build the rest of the paper on.

2.2 Why inelastic demand makes markets moveable

The efficient-market intuition rests on a hidden assumption: that the aggregate demand curve for equities is nearly flat, so that price-insensitive buying or selling is absorbed by an elastic supply of arbitrage capital at essentially unchanged prices. The *inelastic markets hypothesis* argues that this assumption is empirically false at the level of the aggregate market (Gabaix & Koijen 2021). Using flows into and out of the equity market, the authors estimate an aggregate *multiplier* of roughly five: a dollar of price-insensitive net purchase of equities raises the aggregate market value of equities by about five dollars. The mechanism is that most holders — index funds, mandated allocators, buy-and-hold households — do not sell into a price rise, so the residual pool of investors who *will* adjust their equity share in response to price is small, and accommodating a forced flow requires a large price move to coax them.

The arithmetic of the multiplier is worth spelling out, because it is where the intuition breaks. Under the flat-demand null, a forced net purchase of, say, \$100 billion of equities is met by an equally elastic supply and moves aggregate value by essentially nothing — the arbitrage community sells \$100 billion of its holdings at unchanged prices, absorbing the flow. Under an inelastic demand curve with a multiplier of five, that same \$100 billion *raises aggregate equity value by roughly \$500 billion*, because the pool of investors willing to sell into the rise is small and must be tempted with a materially higher price. The gap between "\$0" and "\$500 billion" is the entire distance between the efficient-market reflex and the flow-based view, and the evidence sides with the latter at the aggregate level. The companion framework, *demand-system asset pricing*, formalizes this by modelling each investor's holdings as a function of prices and characteristics and clearing the market for the price vector, rather than assuming a representative agent with a flat demand curve; estimated at the individual-stock level, its demand elasticities are far lower than a frictionless model predicts (Kojien & Yogo 2019). Its practical upshot for this paper is a licence to take flows seriously: if demand is inelastic, then a forced seller of a given size has a *predictable* first-order effect on price, and the interesting questions become who the forced traders are, how large their positions are, and when their triggers fire — exactly the questions the rest of this paper asks of dealers, volatility-target funds, CTAs, and calendar programs.

Two cautions travel with the hypothesis, and honesty requires stating them. The multiplier is estimated at the *aggregate market* level and over quarters; it is not a licence to claim that any particular intraday flow of a known dollar size will move the index by five times its notional. And the estimate is itself contested and identification-dependent — it relies on instruments for the price-insensitive component of flow, and reasonable economists disagree about their validity. What the literature establishes is *plausibility*, tagged: that forced demand is a first-order driver of prices, not a rounding error competed instantly to zero. That plausibility is the foundation the map is built on; the map itself never claims more than a disciplined prior.

2.3 The four families, previewed

The rest of the paper treats four flow families, and it is worth naming them once, up front, with their coordinates on the taxonomy, so the reader carries the whole map into the detailed sections. Each is a row of the grid in Figure 1, read together with its position on the elasticity axis:

- **Dealer hedging (Section 4)** — trigger: price (gamma), time (charm), and volatility (vanna). Elasticity: *either end*, set by the gamma sign — a long-gamma book is elastic and damping, a short-gamma book is inelastic and amplifying. This dual nature makes it the switch that governs the others.
- **Volatility-target and risk-parity (Section 5)** — trigger: volatility. Elasticity: inelastic, pro-cyclical — sells into rising realized volatility, buys into calm. A pure mechanical amplifier with the same sign as a short-gamma dealer.
- **CTA trend-following (Section 6)** — trigger: price (an established trend). Elasticity: inelastic at the moment of a flip, when the position must reverse; otherwise slow. Reinforces or cancels the dealer flow depending on the gamma sign.
- **Passive and calendar (Section 7)** — trigger: the clock. Elasticity: the most inelastic of all (a date-driven rebalance is indifferent to price) but also the most *anticipated*, so its footprint is largely front-run away.

Two families are volatility-triggered amplifiers (vol-target, and the short-gamma side of dealer hedging), one is price-triggered and regime-dependent (CTA, and the gamma side of dealer hedging), and one is clock-triggered and heavily arbitrated (calendar). Holding these four coordinates in mind is enough to read Section 8's regime map before any of the detail arrives: the dangerous regimes are the ones in which the price- and volatility-triggered amplifiers all point the same way, and the benign ones are the ones in which the dealer switch is set to *damp*. The intervening sections give each family its sign, size, and timing so that claim can be made precisely.

3 Price impact — the common language of all flows

If flows move price, then to compare a dealer gamma hedge against a risk-parity rebalance we need one number that both produce: the *price impact* of a given quantity traded. This section develops that number. It is the common currency of the whole paper, and getting its shape right — in particular, that impact is *concave* in size, not linear — changes how one thinks about small flows.

3.1 Temporary versus permanent impact

When a participant trades a quantity Q of an instrument whose average daily volume is V , the price responds in two distinct ways, and the distinction is not pedantic — it decides whether a flow leaves a lasting mark or merely a transient dent.

Temporary impact is the concession the trader pays for demanding immediacy: to fill quickly they must cross the spread and walk up (or down) the book, and the price relaxes back once the pressure is removed. Temporary impact is a cost to the initiator and a rent to the liquidity provider; it decays over minutes to hours as the book refills (Bouchaud, Farmer & Lillo 2009). *Permanent impact* is the part of the move that does not relax — the market's revised belief about value, updated because the trade itself was read as information. A metaorder (a large parent order sliced into many child executions) leaves a characteristic signature: price drifts in the direction of the order while it executes, then partially reverts after it completes, settling at a permanent level above (or below) the start.

For the flows in this paper the distinction maps cleanly onto the elasticity axis of Section 2. A *liquidity-supplying* elastic flow — a long-gamma dealer fading a move, a value buyer stepping in — largely earns temporary impact rather than paying it. A *liquidity-demanding* inelastic flow — a short-gamma dealer chasing, a target-vol fund deleveraging into weakness — pays temporary impact and, because its trading is correlated with others doing the same thing at the same time, can also drive a semi-permanent move by exhausting the standing book. The most dangerous configurations in Section 8 are precisely those in which several inelastic flows demand liquidity in the same direction at once.

3.2 The square-root law

The central empirical regularity of price impact is that it is *concave* in size: the impact of a metaorder grows roughly as the *square root* of the quantity traded, not linearly. In the standard form,

$$\text{impact} \approx Y \cdot \sigma \cdot \sqrt{(Q / V)},$$

where σ is the instrument's daily volatility, Q/V is *participation* (order size as a fraction of daily volume), and Y is a dimensionless prefactor of order one-half to one. This *square-root law* is one of the most robust stylized facts in microstructure: it holds across markets, asset classes, and epochs, for metaorders spanning several orders of magnitude in size, with Y remarkably stable (Tóth et al. 2011); direct estimation on equity data finds a power close to 0.6, consistent with the square root (Almgren et al. 2005).

The practical content of concavity is easiest to see by contrast with the linear intuition most people carry — that trading twice as much should cost twice as much. Figure 2 draws the square-root law against a linear benchmark calibrated to agree with it at 16% participation, with mid-range literature parameters ($\sigma = 1\%$ daily, $Y = 0.7$, band $Y \in [0.5, 1]$).

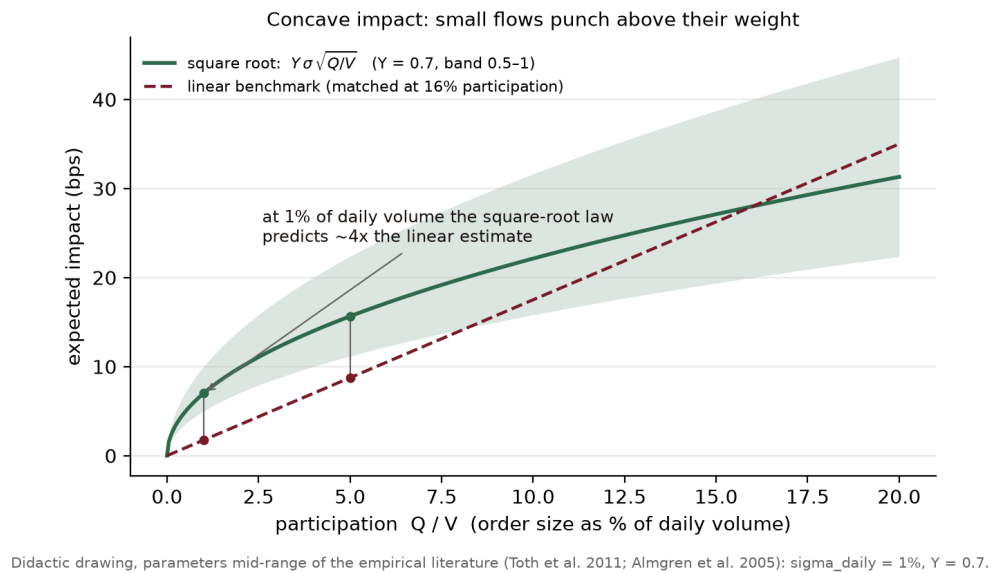


Figure 2. The square-root impact law against the linear intuition, drawn with mid-range literature parameters (daily volatility 1%, prefactor $Y = 0.7$, band 0.5 to 1). The linear benchmark is calibrated to agree with the square-root law at 16% participation, which exposes the disagreement at small sizes: at 1% of daily volume the square-root law predicts about 7.0 bps of impact against roughly 1.8 bps for the linear estimate, a factor of about four. The concave law says small flows punch far above their notional weight. Reproduce with `figures/fig_sqrt_impact.py`.

The numbers behind the figure (`figures/fig_sqrt_impact.py`): at 1% of daily volume the square-root law predicts about 7.0 bps of impact (literature band 5.0–10.0 bps) versus about 1.8 bps for the calibrated linear line — the square-root estimate is roughly *four times* larger at that size. At 0.1% participation the gap widens to more than tenfold (2.2 bps versus 0.2 bps); by 20% participation the two laws have crossed and the linear line predicts *more* (35 bps versus 31 bps). The lesson is the one concavity always teaches: impact per unit size is *largest for the smallest orders*. A modest, price-insensitive flow that a linear model would dismiss as negligible can move price meaningfully — which is exactly why a steady drip of forced hedging or rebalancing, small on any single print, adds up to a first-order driver.

3.3 Why the law is a square root, not a line

Two distinct models of impact circulate, and the difference between them is not cosmetic. The older, tractable one is the *linear* model associated with Kyle: an informed trader's demand is absorbed by market makers who

cannot distinguish it from noise, and equilibrium price impact is linear in signed order flow, with a slope (Kyle's λ) that measures illiquidity. Linear impact is analytically convenient and is a reasonable description of the *instantaneous* response to a single small market order. But it is the wrong description of a *metaorder* — a large parent order worked over minutes to days — and metaorders are what every flow in this paper actually is.

The empirical square-root law says that the *total* impact of executing a metaorder of size Q scales as $\sqrt{Q}$, and the modern explanation for that shape is a statement about *latent liquidity* (Tóth et al. 2011). The visible order book at any instant is a tiny fraction of the liquidity that will eventually appear; most of the liquidity that absorbs a large order is *latent*, revealed only as the price moves toward it. If the density of this latent liquidity *vanishes* near the current price — a V-shaped profile, thin at the touch and thickening as one moves away — then integrating the order through the thin region produces exactly a square-root relationship between quantity and impact. The concavity is therefore not a curiosity; it is the fingerprint of a market whose near-price liquidity is critically thin, which is why the law is so robust across venues and why it *steepens* precisely when the book is stressed and the latent liquidity has withdrawn.

This also clarifies the temporary/permanent split of Section 3.1. As a metaorder executes, it pushes price along the concave curve (the peak impact, at completion, is the $\sqrt{Q}$ value). After it stops, the price *reverts* part way, because the temporary component — the liquidity concession — refills, leaving a permanent residue that is a fraction (empirically often around two-thirds) of the peak. A flow that is *anticipated* by others, as most calendar flows are, sees its temporary component partly pre-empted: liquidity providers position ahead of it, so the initiator pays less concession but the anticipation itself moves price before the flow arrives. This is the mechanism by which a known, dated, inelastic flow (a Russell reconstitution, a turn-of-month rebalance) is largely *front-run into the pre-event window* — a fact Section 7 will need.

3.4 A worked translation: from a dollar flow to a move

To see impact working as a common unit, take a concrete, deliberately round example. Suppose a forced seller must sell \$4 billion of an equity index whose futures and cash together trade about \$200 billion per day, on a day of typical 1%-daily volatility. Participation is $Q/V = 4/200 = 2\%$. The square-root law with the mid-range prefactor gives an impact of roughly $Y \cdot \sigma \cdot \sqrt{(Q/V)} = 0.7 \times 100 \text{ bps} \times \sqrt{0.02} \approx 9.9 \text{ bps}$, with a literature band ($Y \in [0.5, 1]$) of about 7 to 14 bps. That is the *peak* pressure while the program works; the permanent residue is perhaps two-thirds of it, on the order of 6–7 bps. A linear model calibrated at 16% participation would have predicted only about 3.5 bps for the same order and would have badly understated the move.

The example makes three points that the rest of the paper leans on. First, single-digit-basis-point moves *per forced flow* are the right order of magnitude — individually small, which is why they are easy to dismiss, and collectively decisive when several align. Second, the translation runs both ways: observing an unexplained 10 bps move on a quiet news day and inverting the law gives an implied metaorder of roughly 2% participation, a rough but disciplined way to ask "how big was the flow that did this?" Third, the calibration is *soft* — the band spans a factor of two, and in a stressed regime the effective Y is larger still — so impact estimates are order-of-magnitude tools for comparing flows, never precise predictions. With that unit in hand, Sections 4–7 can each report their flow as a participation figure and let impact convert it to a move.

3.5 Why impact is the right unit

Impact is the common currency because it is *dimensionless in the right way*: it converts a quantity, expressed as participation Q/V , into a price move in basis points, using only the instrument's own volatility as a scale. That lets otherwise incomparable flows be laid side by side. A dealer desk that must buy, say, 5% of a day's futures volume to re-hedge a short-gamma book, and a risk-parity complex that must sell 3% of the day's volume to cut risk after a volatility spike, are no longer apples and oranges: the first is worth about 16 bps of upward pressure and the second about 12 bps of downward pressure, on the same 1%-daily-vol scale, and the *net* is what the tape shows. The point is not that these participation figures are known precisely — they are not, and Section 8 is candid about that — but that once each flow is expressed in participation, impact gives them a single ruler.

Three caveats keep the ruler honest. First, the square-root law is a statement about *metaorders* — coordinated parent orders worked over time — not about a single market order, and the flows here are best thought of as slow metaorders. Second, the prefactor Y is genuinely uncertain (the band in Figure 2 is not decoration), so impact estimates are order-of-magnitude, not precise. Third, impact is *regime-dependent*: in a thin, stressed book the same participation costs far more than the calm-market calibration suggests, and the concave curve steepens — which is why the same forced flow that is a gentle nudge in a calm regime becomes a cascade in a stressed one. With those caveats stated, we can now measure each flow family in this unit.

4 Dealer hedging flow (embedded recap of the dealer-flows-and-GEX paper)

Dealer hedging earns its own flagship treatment in the dealer-flows-and-GEX paper, where the estimation mechanics — how one infers dealer positioning from the option chain, and the assumptions that inference rests on — are developed in full. This section is deliberately narrower. It recaps the *mechanism* just enough to state dealer re-hedging as a flow with a sign, a size, and a timing, and then asks the question that belongs to the map rather than to the dealer-flows-and-GEX paper: *how does this flow stack against the others?* The canonical description of the mechanism lives in the author's private research note on market-maker hedging and is not paraphrased-duplicated here; the recap below is the public, mechanism-only version.

4.1 The mechanism, in one paragraph

Market makers take the other side of customer option orders and do not want the directional risk, so they *neutralize their delta* by trading the underlying. Because an option's delta is not constant, the hedge is not a one-time trade: as the world moves, delta drifts and the dealer must re-trade the underlying to stay neutral. Delta drifts through exactly three channels, and each is a Greek: *gamma* (delta changes when the underlying *price* moves), *charm* (delta changes with the passage of *time*, deterministically, and explosively near expiry), and *vanna* (delta changes when *implied volatility* moves). Aggregated across the whole dealer book and across thousands of contracts, these mechanical re-hedges become a measurable force in the underlying — a flow. The single most consequential property of that flow is its *sign*, which flips with the sign of the dealer's aggregate gamma.

4.2 Sign, size, and timing

Sign. When dealers are net *long* gamma (the aggregate gamma-exposure measure is positive), a price rise increases their delta and they *sell*; a price fall decreases their delta and they *buy*. This is *anti-cyclical* — sell rallies, buy dips — so it supplies liquidity and *damps* realized volatility. This is the mechanism behind the author's hypothesis H001, which is formulated on external Tier-B/A evidence and not yet independently validated. When dealers are net *short* gamma (negative exposure), the signs invert: a price rise forces buying and a price fall forces selling. This is *pro-cyclical* — buy strength, sell weakness — so it withdraws liquidity and *amplifies* moves, with the possibility of a feedback loop, the content of hypothesis H002 (the best-supported of the four, on Tier-A evidence, still awaiting own validation): positive aggregate dealer gamma is associated with intraday reversal and negative gamma with momentum (Barbon & Buraschi 2021). The charm channel produces a deterministic, time-of-day flow concentrated in the final hours before expiry, whose direction depends on the open-interest distribution and whose footprint is *pinning* toward high-gamma strikes — hypothesis H003, conditional on the gamma sign; stock prices are documented to cluster at option strikes on expiration dates, with market-maker re-hedging identified as a cause (Ni, Pearson & Poteshman 2005). The vanna channel produces an IV-driven flow — falling IV under a typical index-hedging posture generates systematic buying, the so-called vanna rally — hypothesis H004, the hardest to identify because vanna is invisible on the price chart and requires a clean IV series.

The direction check earns its keep here. It is easy to state "large gamma before expiry raises volatility"; the mechanism says the opposite for a *long*-gamma book, and an earlier version of the author's own note recorded exactly that sign error before it was caught. The sign of a dealer flow is a claim to be checked against the mechanism every time, not assumed.

Size. Sizing the dealer flow is genuinely hard, and it is the subject of the dealer-flows-and-GEX paper, not this recap. The honest public statement is threefold. The sign of aggregate gamma is *estimated*, not observed, because it depends on an assumption about who holds which side of each option — and in the 0DTE era, where customers buy both calls and puts in size, that assumption is fragile. The *magnitude* of the re-hedging flow scales with aggregate gamma and with the size of the price move, but converting it to a participation figure requires the estimation apparatus of the dealer-flows-and-GEX paper. And dealers hedge in *discrete bands*, not continuously, so the flow is lumpier and more threshold-driven than the smooth mechanism suggests. What the map needs from the dealer-flows-and-GEX paper is only the order of magnitude and, above all, the sign; the precise level is that paper's problem and is never published here as a licensed number.

Timing. The three channels differ in *when* they fire, and this is what lets them be placed on the flow calendar of Section 8. Gamma re-hedging is *continuous* and price-triggered — loudest when the underlying is near a large-gamma strike and near expiry. Charm is *time-triggered* and concentrated in the last one to two hours of the session, sharpest on expiry days. Vanna is *volatility-triggered* and fires whenever IV moves materially, which clusters around stress and its aftermath.

The three channels and their signs are worth tabulating, because the sign is the whole game and it is easy to get wrong:

Channel	Trigger		Long-gamma / typical index book	Short-gamma book	Footprint
Gamma	underlying moves	price	sell rallies, buy dips (damps)	buy rallies, sell dips (amplifies)	narrow ranges vs. cascades
Charm	passage of time		hedge toward pin as delta decays	hedge away from pin	end-of-day drift, expiry pinning
Vanna	implied moves	volatility	falling IV → buy (vanna rally)	falling IV → sell	post-stress bid or fade

The table is a compression of the canonical mechanism and of the four hypotheses; each row is a claim with a sign, and each sign is contingent on the dealer-positioning assumption that the estimation apparatus of the dealer-flows-and-GEX paper exists to test.

4.3 An order-of-magnitude size, in the impact unit

Because the dealer-flows-and-GEX paper owns the estimation, the map needs only a sense of scale, and Section 3 supplies the ruler. A back-of-envelope shows the dealer flow is the same order of magnitude as the other flows, not larger and not negligible. Suppose the aggregate dealer gamma is such that a 1% move in the index changes the desk's aggregate delta by an amount requiring, say, 3% of the day's index-futures volume to re-hedge — a figure consistent with the sizes practitioners discuss on large-gamma expiry days, offered here only as an order of magnitude. By the square-root law at 1%-daily volatility, 3% participation is worth about $Y \cdot \sigma \cdot \sqrt{0.03} \approx 0.7 \times 100 \text{ bps} \times 0.173 \approx 12 \text{ bps}$ of pressure per 1% move re-hedged. If the desk is *long* gamma that 12 bps leans *against* the move (a damping ceiling and floor); if it is *short* gamma the same 12 bps leans *with* it (accelerant). The number is soft — the participation is an assumption and the prefactor a band — but it establishes the point the map requires: a dealer re-hedge and a volatility-target rebalance and a CTA flip are all *single-digit-to-low-double-digit basis-point* flows on a normal day, comparable in size, so which one dominates is decided by *alignment and regime*, not by one flow dwarfing the others. Only in stress, when several align and the impact curve steepens, does the total become large.

4.4 The 0DTE complication

Any honest recap must flag the way the ground has shifted since 2022. The explosive growth of *zero-days-to-expiry* (0DTE) index options has changed both the size and the sign-reliability of the dealer flow. On the size side, a large 0DTE open interest concentrates enormous gamma and charm into a single session, so the intraday re-hedging flow and the end-of-day charm flow are larger and more compressed than the pre-0DTE era implied — the gamma "pin" and the late-day drift are sharper. On the sign side, 0DTE has made the dealer-positioning assumption more fragile than ever: in the pre-0DTE world one could lean on the stylized fact that customers are net buyers of downside puts (leaving dealers short puts, long the crash), but 0DTE customer flow buys both calls and puts in size and intraday, so the *sign* of aggregate dealer gamma can be genuinely ambiguous and can flip within a session. This is why the hypotheses of Section 4.2 are all formulated rather than supported, why their tests demand intraday chain data rather than end-of-day snapshots, and why the map treats the dealer sign as an *estimate to be gated*, not a known input. It is also why

the regime gate of Section 8 — which reads stress from the volatility complex directly — is a valuable cross-check on a gamma sign that the option chain alone may be mis-stating.

4.5 How it stacks against everything else

Placed on the elasticity axis of Section 2, the dealer flow is remarkable for being able to sit at *either end* depending on the gamma sign, and this is the property that makes it the linchpin of the regime map. A *long*-gamma dealer is an elastic, liquidity-supplying, volatility-damping presence — functionally on the same side as a value buyer, leaning against moves. A *short*-gamma dealer is an inelastic, liquidity-demanding, volatility-amplifying presence — functionally on the same side as a target-vol fund forced to deleverage, leaning *with* moves.

That dual nature sets up the central interaction of the whole paper, developed fully in Section 8. In a calm, positive-gamma regime the dealer flow is a *stabilizer*, and it tends to cancel or absorb the other flows: it buys the dips that a CTA might be selling, and it caps the rallies that momentum might be chasing, so the tape churns. In a stressed, negative-gamma regime the dealer flow *reverses polarity and aligns* with the other inelastic sellers: it sells weakness at the same time a volatility-target fund is deleveraging and a trend-follower is flipping short, and the three forced sellers demand liquidity in the same direction from a book that is simultaneously thinning — the mechanical anatomy of a cascade. Dealer hedging, in other words, is not just one flow among four; it is the *switch* that decides whether the other flows are damped or reinforced.

There is one further asymmetry that elevates the dealer flow above its peers on the map. The other three families are *slow* relative to it: a vol-target rebalance reacts to trailing realized volatility over days, a CTA flip needs a multi-month return to invert, and a calendar flow fires on a fixed date. The dealer gamma flow, by contrast, is *continuous and intraday* — it responds to every tick, within the session, in real time. That makes it the flow most capable of turning a slow-building imbalance into a fast move: when the slow amplifiers (vol-target, CTA) have quietly tilted the book to one side and a short-gamma dealer sits underneath, a single catalyst can trigger a re-hedging cascade that plays out in hours, not days. The dealer flow is therefore both the switch *and* the accelerator — the reason a regime that looks stable on a daily chart can break intraday. The rest of the paper measures the other three so that, in Section 8, the switch can be read, and the gate of Section 8.3 exists precisely to catch the intraday stress the slow flows cannot.

5 Volatility-target & risk-parity flows

The first of the three non-dealer flow families is the one most transparently mechanical, and therefore the easiest to reconstruct honestly on free data: the deleveraging of *volatility-target* and *risk-parity* strategies. These strategies share a single reflex — hold risk, not capital, constant — and that reflex makes them sell into rising realized volatility and buy back into calm, a pro-cyclical amplifier with the same sign as a short-gamma dealer.

5.1 The mechanism

A volatility-target fund runs a simple control law. Let the target annualized volatility be σ^* and the estimate of the asset's recent realized volatility be RV_t . The rule sets the allocation to

$$w_t = \sigma^* / RV_t \text{ (capped at some maximum leverage),}$$

so that the portfolio's forecast volatility, $w_t \times RV_t$, equals the constant target σ^* . *The mechanical consequence is immediate: because w_t is inversely proportional to realized volatility, a rise in RV forces w down — the fund sells — and a fall in RV lets w rise — the fund buys back. The size of the forced trade is the change in the allocation, $\Delta w_t = \sigma^*(1/RV_t - 1/RV_{t-1})$, which is largest exactly when RV is *changing* fastest, i.e. in the first days of a volatility spike. Risk parity generalizes the same reflex across several assets, sizing each inversely to its own risk (and often scaling the whole book to a portfolio-volatility target and accounting for cross-asset correlations), but the equity leg behaves identically: cut equities when equity volatility spikes. Because realized volatility rises precisely when prices fall sharply, the control law *sells into weakness and buys into strength* — the definition of pro-cyclical, liquidity-demanding, inelastic flow. It is a mechanical amplifier, and it is *synchronized* across the industry because every fund's risk model reacts to the same volatility spike at roughly the same time, which is what turns a diffuse set of private rules into a correlated market-wide flow.*

5.2 A transparent free-data reconstruction

To make the mechanism concrete without any licensed data, Figure 3 applies *one* pre-committed rule to daily S&P 500 closes (public Yahoo Finance chart API, cached). The rule, fixed in advance and not tuned: target 10% annualized volatility; estimate realized volatility as the annualized standard deviation of the trailing 21 trading days; set allocation = $\min(1, 10\% \div \text{trailing-realized-vol})$, long-only and capped at fully invested; hold the allocation decided at today's close on the next day (no look-ahead); and read the day's rebalancing flow as the change in allocation, in percent of the fund's net asset value. This is illustrative of the *mechanism*; it is not a calibrated model of any real book.

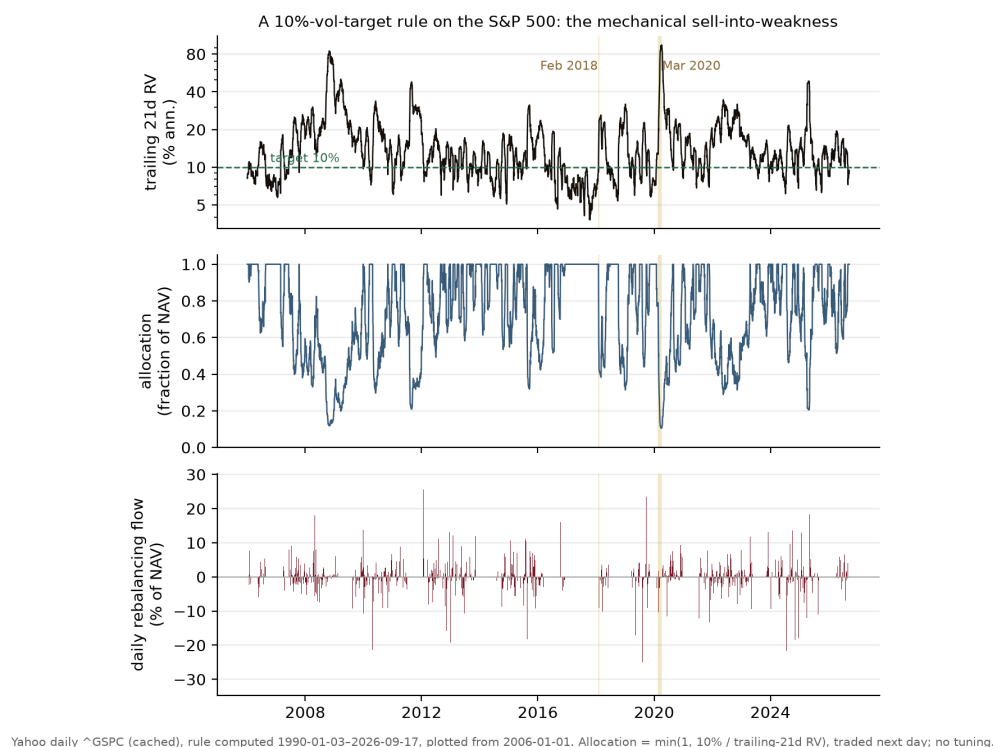


Figure 3. A 10%-volatility-target rule on the S&P 500, computed on daily closes from 1990 to 2026 and plotted from 2006. Top: trailing 21-day realized volatility against the 10% target, log scale. Middle: the implied allocation, capped at fully invested, which spends most of its time below one and collapses in stress. Bottom: the daily rebalancing flow in percent of NAV, with the February 2018 and March 2020 deleveraging episodes shaded. The rule is fixed in advance and traded next day, with no tuning. Reproduce with `figures/fig_voltarget.py`.

The printed numbers (`figures/fig_voltarget.py`, sample 1990–2026, 9,244 daily returns) tell the story the mechanism predicts. The average allocation is 0.74 — a 10%-target fund is *usually deleveraged*, because equity realized volatility usually exceeds 10% — and it ends the sample fully invested. The flow is heavy: the mean absolute daily rebalance is 1.63% of NAV, which annualizes to roughly 411% of NAV in one-way turnover, and on 841 days in the sample the fund had to move more than 5% of NAV in a single day. The stress episodes are where the amplifier bites. In the *February 2018* volatility spike ("Volmageddon"), the rule cut allocation by 31.7% of NAV in a *single day* (5 February, allocation 0.865 → 0.548) and by a cumulative 59% of NAV over the thirteen-day episode. In *March 2020* the deleveraging ran longer and deeper: a cumulative 68% of NAV cut over 35 days, with the allocation bottoming at 0.11 — 89% deleveraged — on 6 April 2020. The single largest one-day deleverage in the whole sample was 34.3% of NAV, in November 1991 (outside the plotted window). That the largest single-day cut in a 36-year sample sits in an unremarkable month — not in October 2008 or March 2020 — is a warning in plain sight: these one-day magnitudes are inflated by the *equal-weight* 21-day window, which steps discretely as a large return enters and, twenty-one sessions later, exits the window. An exponentially-weighted or blended estimator, which most real vol-target books use, would smooth those steps and shrink the single-day figures, so the 31.7% is illustrative of the mechanism, not a calibrated estimate of any fund's one-day flow. These are the mechanical sell orders that arrive *because volatility rose*, regardless of price, and they arrive in size precisely when the book is thinnest.

5.3 What a deleveraging day is worth, and why the two cascades differ

The reconstruction gives an allocation change per dollar of AUM; the impact unit of Section 3 converts it to a move. The conversion requires an estimate of the sensitive AUM, which is exactly the quantity Section 5.4 warns is soft, so the arithmetic is illustrative. Suppose the volatility-target and risk-parity complex sensitive to a spike controls \$300 billion of equity exposure — a middle-of-the-road figure among published desk estimates. A one-day cut of 31.7% of NAV, as on 5 February 2018, then implies roughly \$95 billion of index selling compressed into a session. Against a \$200-billion-per-day market that is nearly 50% participation, deep in the region where the impact curve is steep and the calm-market calibration has broken down. Even spreading the sale over the thirteen-day episode leaves several percent of daily volume of forced selling every day for two weeks — a persistent, one-directional, price-insensitive drip precisely of the kind Section 3 showed to be a first-order driver. The point is not the exact dollar figure, which no public dataset pins down, but the *scale*: in a genuine deleveraging the vol-target flow is not a single-digit-basis-point nudge like a normal-day dealer hedge, it is a market-moving flood.

The two shaded episodes differ in a way the mechanism explains, and the difference matters for the regime map. February 2018 was a *fast* event: a volatility spike concentrated in days, with the deleveraging front-loaded (one 31.7%-of-NAV day) and the recovery relatively quick. March 2020 was a *deep* and *prolonged* one: the deleveraging accumulated to 68% of NAV over five weeks and the allocation reached its 0.11 trough only in early April, weeks after the price low, because trailing realized volatility stayed elevated long after prices stabilized. This lag is a general feature of the flow: because the trigger is *trailing* realized volatility, the buying-back (the re-leveraging) is *slow*, dragging the vol-target bid well into the recovery — a mechanical tailwind that persists long after the panic has passed, and one the map should expect on the right-hand side of any large drawdown.

5.4 Honest caveats

Three caveats keep this from over-claiming, and they are all. First, the *assets under management* and the *trigger levels* of the real volatility-target and risk-parity industry are estimates, not observed quantities; published desk estimates of the total sensitive AUM vary by large factors, and the aggregate flow the industry actually generates on a given day cannot be computed from public data with any precision. Second, *real books differ* in every parameter this reconstruction fixed: the target volatility, the length and weighting of the realized-vol window (many use exponential weighting or blend multiple horizons), the presence of leverage caps and floors, and the speed of rebalancing (some spread the adjustment over days to reduce impact). A single 21-day, 10%-target, next-day rule is a caricature chosen for transparency, not a model of the marginal fund. Third, the *direction of causation* is entangled: the same volatility spike that forces the deleveraging is partly *caused* by the deleveraging, so the reconstructed flow is not an exogenous shock one can cleanly attribute. What the reconstruction honestly establishes is the *shape and sign* of the flow — large, pro-cyclical, concentrated in stress — not its calibrated magnitude. The literature supports the shape: systematic volatility targeting materially changes the drawdown and higher-moment profile of an equity allocation, improving risk-adjusted returns in normal times while contributing to synchronized deleveraging in stress (Harvey et al. 2018).

Despite the caveats, the reconstruction earns its place on the map for a reason the caveats do not touch: its *timing and direction are not estimated at all* — they are dictated by the control law and by the arithmetic of

trailing volatility, and both are fully observable on free data. We may not know the industry's AUM, but we know with certainty that a volatility-target complex of *any* size sells on the days the reconstructed allocation falls and buys on the days it rises, and Figure 3 dates those days exactly. That is precisely the kind of input the regime map needs: not a calibrated flow forecast, but a reliable answer to "which way is the mechanical volatility flow leaning today, and is it accelerating?" The magnitude is a free parameter the map leaves open and estimation in the dealer-flows-and-GEX paper or a dedicated study must pin; the sign and the timing the reconstruction delivers for free, and they are what Section 8 consumes.

6 CTA / trend-following flows

The second non-dealer family is *CTA trend-following*. Managed-futures funds and trend programs hold long positions in uptrending markets and short positions in downtrending ones, sized by volatility, and they *flip* — forced to buy or sell — when a trend reverses. Their positioning is estimable from public prices because the signals they use are, by design, simple and public.

6.1 The mechanism and the proxy

The canonical trend signal is *time-series momentum*: the sign of an asset's own trailing return over a lookback horizon predicts its next-period return, and a diversified portfolio that goes long past winners and short past losers across many markets has earned a persistent premium (Moskowitz, Ooi & Pedersen 2012). Real CTAs blend many horizons and markets, but the core positioning signal is well proxied by a small ensemble of trailing-return signs. Figure 4 uses *one* pre-committed, standard ensemble with no tuning: the signal is the average of the signs of the trailing 3-month, 6-month, and 12-month returns (63, 126, and 252 trading days), which takes values in $\{-1, -1/3, +1/3, +1\}$. The aggregate sign is the positioning proxy — net long when positive, net short when negative — and a *flip* is a change in that sign. The proxy is computed on the S&P 500 and on the 20-year-plus Treasury ETF (TLT), both from cached free daily data.



Figure 4. A trend-ensemble positioning proxy: the mean of the signs of the trailing 3-, 6-, and 12-month return, computed on the S&P 500 and on long-dated Treasuries (TLT) and plotted from 2004. Green fill marks net-long positioning, red net-short. The proxy is the textbook time-series-momentum family with no tuning; a flip is a change in the aggregate sign. Reproduce with `figures/fig_cta_proxy.py`.

6.2 The current reading and the flip history

The printed numbers (`figures/fig_cta_proxy.py`) give a snapshot and a history. As of 17 September 2026 the S&P 500 proxy reads +1.00 — trend-followers are, on this proxy, *fully long* equities, all three horizons pointing up — while the Treasury proxy reads −1.00, fully short duration. Over the sample the equity proxy is net long 74% of days, the natural consequence of an asset with a positive drift. The flips are infrequent but sharp: filtering to sign changes that persist at least ten trading days, the equity proxy has flipped 32 times over the sample, and its recent flips cluster around the obvious turning points — short on 26 March 2025, back long on 2 June 2025, short again on 20 March 2026, and long once more on 8 April 2026. Each flip is a *forced* trade: at the moment the ensemble sign changes, a trend program that follows it must reverse its entire position, buying or selling in size into a market that has, by construction, just moved a long way. The Treasury proxy flips more often (48 persistent episodes), reflecting the choppier, less persistently-trending character of bond returns over the period.

6.3 Sizing, capacity, and why flips are the dangerous moment

A trend signal is only half the flow; the other half is *sizing*, and CTAs size the same way vol-target funds do — inversely to volatility. A trend program typically holds a position scaled so that each market contributes a target risk, $\text{position} \propto \text{signal} \times (\sigma \backslash RV)$. *Two consequences follow. First, the CTA flow inherits the vol-target flow's pro-cyclicality on top of its trend logic: even without a flip, a volatility spike shrinks the position, forcing selling of a long into weakness. Second, and more important for the map, the flip is the moment of maximum forced trading. Away from a flip the signal is saturated at ± 1 and the position drifts only with volatility; at a flip the signal swings from (say) $+1$ to -1 , so the program must sell its entire long and* establish a short — a position change of twice the book — into a market that, by the construction of a trend*

signal, has just moved far enough to invert a multi-month return. This is why Section 6.2 emphasized the flip dates: they are not incidental annotations, they are the timestamps of the largest CTA orders.

A rough notion makes the scale concrete. Industry trend-following and managed-futures capital is often put in the several-hundred-billion-dollar range, with the equity-index sleeve a fraction of that. If, say, \$150 billion of trend capital is positioned in equity indices and a flip reverses even half of it over a few days, that is \$75 billion of one-directional forced flow — again a multi-percent-of-volume drip, again comparable to the other families, again decisive mainly when it *aligns* with them. And CTA flow carries a distinctive *capacity and crowding* caveat: because the signals are public and simple, the flow is the most anticipatable of the mechanical families after the calendar, so its edge is partly arbitrated and its impact partly front-run — the same erosion Section 7 documents for calendar flows. The proxy of Section 6.2 estimates *positioning*, which is real and forced; it does not estimate a tradable *edge*, which crowding may have competed away.

6.4 Where CTA and dealer flows reinforce versus cancel

The map's question is how this flow interacts with the dealer flow, and the answer depends — again — on the dealer's gamma sign. Consider a sharp equity selloff. The CTA proxy, if the move is large enough to flip the ensemble, turns *seller* just as the trend breaks. If dealers are *long* gamma, they are simultaneously *buying* the dip, and the two flows partly *cancel*: the dealer supplies the liquidity the flipping CTA demands, and the selloff is cushioned. If dealers are *short* gamma, they are *selling* the same weakness, and the two flows *reinforce*: forced CTA selling and forced dealer selling stack on the same side of a thinning book, and the move extends. This is the same switch identified in Section 4.5, now with a second inelastic seller wired into it. The volatility-target flow of Section 5 is the third. The reason Section 8 can speak of a "cascade" as a mechanical object rather than a metaphor is that in a negative-gamma regime all three of these forced flows point the same way at once.

The alignment is not hypothetical; it lines up with the episodes already on the page. The two deleveraging cascades of Figure 3 — February 2018 and March 2020 — coincide with the equity proxy's forced turns in Figure 4 (the ensemble flipped short around the late-2018 breakdown and again in March 2020), so in each case the vol-target complex was dumping exposure *while* the trend proxy was flipping short — two of the three inelastic sellers firing together, into the stress where the dealer switch is most likely thrown to short-gamma. The calm-market mirror is just as instructive: on an ordinary −1% day in a positive-gamma regime, the CTA proxy stays saturated long (no flip), the vol-target rule barely moves (trailing RV rises little), and the long-gamma dealer buys the dip — no flow is forced in size and the dip mean-reverts by the close. The same shock, two regimes, opposite outcomes, and the difference is entirely which flows were compelled to trade and whether the dealer leaned with them or against them. That is the mechanical content Section 8 assembles into a map.

7 Passive & systematic calendar flows

The third non-dealer family trades on *dates you can circle in advance*: index reconstitutions, month-end and quarter-end rebalancing by mandated allocators, pension and target-date-fund flows, and corporate buyback windows and their blackout periods. These flows are the purest form of price-inelastic, clock-triggered demand — a pension fund rebalancing to policy weights on the last day of the quarter is indifferent to that

day's price — and they have generated a large folklore. The discipline of this section is to separate the *measurable footprint* from the folklore with a pre-committed test, and to report whatever it shows.

7.1 The calendar-flow mechanisms

Several mechanisms populate the calendar. *Index reconstitutions* force every index fund to buy the added names and sell the deleted ones on the effective date; the annual reconstitution of the broad small-cap benchmarks is the archetype, concentrating tens of billions of dollars of mechanical trading into the closing auction of a single June session. Because the membership changes are published in advance, this flow is the textbook case of Section 3.3's anticipation: much of the impact is front-run into the days before the effective date, and the reconstitution *close* itself is one of the highest-volume prints of the year. *Month-end and quarter-end rebalancing* arises because a portfolio held to fixed policy weights drifts as assets diverge over the month; restoring a 60/40 book after a strong equity month means *selling equities and buying bonds* into month-end, with the mechanical size proportional to the month's cross-asset return spread — a multi-day divergence of several percent can imply tens of billions of dollars of index selling from the mandated-allocator complex alone, and the sign flips cleanly after a weak equity month. *Pension and target-date funds* add a slower, glide-path version of the same rebalancing, concentrated at quarter-ends. *Buyback programs* bid for a company's own shares during open windows and are legally constrained to relative silence during *blackout* periods in the weeks before earnings, so the aggregate corporate bid — one of the largest structural sources of equity demand — switches on and off on a schedule tied to the earnings calendar, thinning predictably in the pre-earnings blackout and returning after. Each of these is a real, dated, inelastic flow with a knowable sign and date; the question the test must answer is whether they aggregate into a *footprint visible in index returns*, or whether anticipation has flattened them below detectability.

7.2 A pre-committed turn-of-month test

The most famous calendar claim is the *turn-of-month* (TOM) effect: the assertion that equity returns are concentrated in a narrow window around the month boundary, driven by the mechanical rebalancing and reinvestment flows above. A classic study found, strikingly, that *all* of the market's excess return over Treasury bills historically accrued in the four days around the turn of the month, with the rest of the month flat on average (McConnell & Xu 2008). Figure 5 subjects this to a pre-committed test on the S&P 500, reported as-is whatever it shows. The window is fixed in advance as the last four trading days plus the first three trading days of each month; the metric is the mean daily close-to-close return, TOM window versus all other days; the sample is 1996 to 2026; and the split is fixed in advance — in-sample 1996–2015, out-of-sample 2016 onward, opened once. Confidence intervals are 95%, normal approximation.

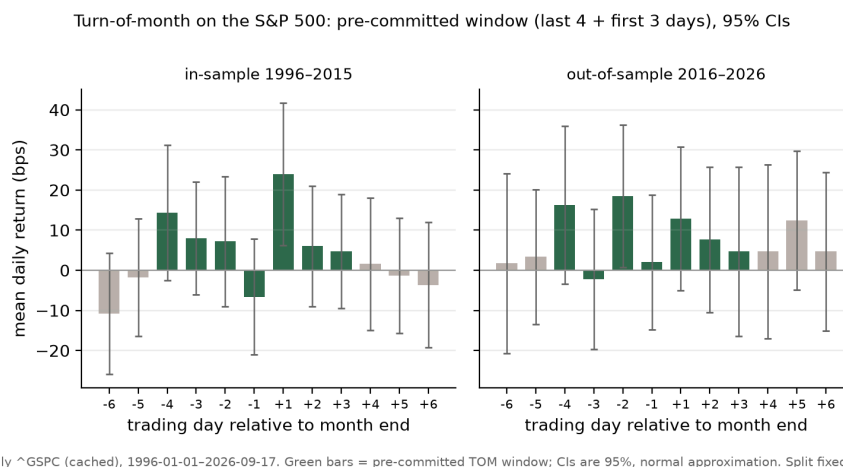


Figure 5. A pre-committed turn-of-month test on the S&P 500, 1996 to 2026. Mean daily return in basis points by trading-day position relative to month end, with 95% confidence intervals, split into in-sample (1996 to 2015) and out-of-sample (2016 onward). Green bars are the pre-committed turn-of-month window (last four and first three trading days); grey bars are the rest of the month. The window, split, and metric were fixed before the data was examined. Reproduce with `figures/fig_turn_of_month.py`.

The result is a textbook case of an effect that is *real in-sample and fragile out-of-sample* (`figures/fig_turn_of_month.py`). In-sample, the turn-of-month window averaged +8.2 bps per day against +0.6 bps for the rest of the month — a gap of +7.6 bps per day, with a *t*-statistic of +2.07, marginally significant, and the sign concentrated as the folklore predicts (the first trading day of the month is the single strongest position, at +23.9 bps in-sample). Out-of-sample, the *sign persisted* — the TOM window still averaged +8.5 bps against +4.0 bps for the rest — but the *gap shrank to +4.5 bps per day with a t-statistic of +1.00*, which does not clear any conventional significance bar. The honest reading is that the turn-of-month footprint is present and correctly signed, but that its edge over the rest of the month attenuated by roughly 40% out-of-sample and lost statistical significance. This is neither a debunking nor a confirmation; it is the characteristic decay of a widely-known calendar anomaly once it is traded, and it is reported here precisely because a pre-committed test is obliged to report the near-null as loudly as it would have reported a triumph. Two per-position cells survive well in both periods — the fourth-to-last trading day (+14.4 bps in-sample, +16.3 bps out-of-sample) is the most stable — but with per-position sample sizes of 129–240 days and confidence intervals of ± 15 –22 bps, no single cell is individually significant, and one should not mine them.

7.3 What the calendar test teaches

The turn-of-month result generalizes to the whole calendar family. These flows are the most *identifiable* of any in the paper — the dates are known exactly, and the mechanical direction is often unambiguous — and yet even here the measured footprint is modest and decays out-of-sample once it is public. The lesson is not that calendar flows are fictional; the mechanism is real and the in-sample sign is right. The lesson is that *known, inelastic, dated flows are the first to be arbitrated toward the edge of detectability*, because they are the easiest to anticipate. A flow you can circle on a calendar is a flow others can circle too, and the front-running that compresses reconstitution impact into the pre-event days is the same force that erodes the tradable turn-of-month edge. Calendar flows belong on the map — they tell you *when* forced rebalancing will

lean on the tape — but their standalone predictive edge is small and shrinking, and the honest use of them is as a *conditioning* variable in Section 8, not a standalone signal.

It is worth separating two things the test does *not* let us conclude. It does not show that the calendar *flow* has weakened — the mechanical rebalancing of a 60/40 book into month-end is as large as ever, growing with the assets under management of the mandated-allocator complex. It shows only that the *return footprint* of that flow, measured as a mean-return premium on index closes, has become statistically indistinguishable from noise out-of-sample. Those are compatible: a large, anticipated flow can move price *before* the dated window (the front-running of Section 3.3) and leave the window itself looking unremarkable, so a test that measures returns *inside* the window will understate a flow that is real but time-shifted. The clean way to test the other calendar mechanisms — reconstitution, quarter-end, buyback blackout — is therefore not a return premium on the event day but an *event study of the pre-event drift and the volume signature*, which requires the intraday and volume data those specific mechanisms live in and is left to a focused study rather than attempted here on daily closes. The honest summary is narrow and defensible: on daily S&P 500 closes, over three decades, with one pre-committed window and split, the turn-of-month return premium is correctly signed, marginal in-sample, and insignificant out-of-sample — and that is exactly as much as a free-data test of this claim can support.

8 Putting it together: a flow calendar & regime map

The preceding sections measured four flow families — dealer hedging, volatility-target deleveraging, CTA trend-following, and calendar rebalancing — each in the common unit of price impact and each with its sign and timing stated. This section assembles them into a *map*: which flow dominates in which regime, how the map combines with the volatility-factor gate from the volatility-surface paper, and — at equal length — why the map is a disciplined prior rather than a validated model.

8.1 The regime map

The organizing insight of Section 4.3 is that the *dealer gamma sign is the switch* that decides whether the other inelastic flows are damped or reinforced. That gives a two-regime backbone, which the author's own daily process labels with four names. Those four names are private labels, not evidence: everything a reader needs to follow — and to test — the argument is the *public* two-regime backbone and the hypotheses stated in full below; the taxonomy and the H-numbers are the author's own shorthand and carry no weight a reader cannot reconstruct from the mechanism and free data.

In a *calm, positive-gamma regime*, dealers are long gamma and lean against moves; volatility-target funds are near fully invested and quiet; the CTA proxy is typically long and not flipping; and calendar flows, if any, tend to *pin* price toward high-gamma strikes into an expiry. Every active flow is either liquidity-supplying or dormant, and the tape churns in a range. In the author's private regime taxonomy this is the *Chop* label when the vanna profile is unsupportive and the *Grind* label when it is supportive of a slow drift — the distinction being whether the IV-driven (vanna) flow adds a gentle directional bias to the volatility-damping (gamma) flow. Realized volatility runs below the implied expected move; mean-reversion setups are favoured; the danger is small.

In a *stressed, negative-gamma regime*, the switch throws. Dealers are short gamma and now sell weakness and buy strength; a volatility spike has forced the volatility-target complex to deleverage (Section 5); and if the move is large enough the CTA proxy flips short and adds forced selling (Section 6). Three inelastic sellers demand liquidity in the same direction from a book that is simultaneously thinning, and impact (Section 3) steepens because the calibration that held in calm markets no longer does. This is the *Expansion* or *Trend* regime; in its most violent form, with the vanna and correlation factors also pointing the wrong way, it is a *Cascade*. Realized volatility exceeds the implied expected move; breakout and continuation dominate; the danger is large and self-reinforcing. The map's single most useful output is the warning that this regime is the one in which *all four flows can align*, and that the alignment is mechanical, not sentimental.

8.2 A regime-to-flow table and a worked flow budget

The map can be compressed into a single table that says, for each regime, which flows are active and on which side, and what the volatility gate should read. It is a prior, not a lookup table for trades:

Regime	Dealer gamma	Vol-target	CTA	Calendar	Gate signature	Net character
Chop	long (damps)	invested, quiet	long, saturated	pins into expiry	low level, contango, low VVIX, flat skew	range, mean- reversion
Grind	long (damps)	invested	long	mild trend into vacuum	low level, contango, supportive vanna	slow directional drift
Expansion / Trend	short (amplifies)	deleveraging	flipping	rebalance flows unreliable	rising level, flattening term	one- directional, extends
Cascade	short (amplifies)	forced deleveraging	flipped short	month-end selling stacks	backwardation, VVIX spike, steep or <i>flat</i> skew	self- reinforcing, gap-prone

Reading down the two right-hand columns is the whole message of the paper: the benign regimes have *one* active pusher (a damping dealer) and dormant others, while the dangerous regimes have *all four* pointing the same way. A worked *flow budget* for a stress day makes the alignment quantitative. Take a -2% index day in a negative-gamma regime. Using the order-of-magnitude participations of the previous sections: a short-gamma dealer re-hedge worth perhaps 3–5% of volume of *selling* into the fall (Section 4.3); a vol-target complex forced to cut several percent of volume as trailing RV jumps (Section 5.3); and, if the move breaks a trend, a CTA flip adding more selling over the following days (Section 6.3). Summed, forced sell-side participation on the order of 10% of daily volume is plausible — and by the *steepened* stress-regime impact curve of Section 3.2, 10% participation is worth not the calm-market ~ 22 bps but potentially several times that, which is how a -2% day becomes a -4% day. Run the same budget in a positive-gamma regime and the dealer term flips to a *buyer* of the dip, partly offsetting the vol-target and CTA selling, and the same shock is absorbed in a -2% day that mean-reverts. The budget is illustrative — every participation in it is an estimate — but it shows *why* the regime, not the shock, sets the outcome.

8.3 Combining with the volatility-surface paper's volatility-factor gate

The regime map says *who is pushing*; the volatility-factor gate from the volatility-surface paper says *which regime you are in*, and the two are meant to be read together. The volatility-surface paper builds an orthogonal set of five volatility factors, each with one non-redundant proxy: the *level* of index-option implied volatility, the *volatility of volatility* (VVIX), the *term-structure slope* (the ratio of a 30-day to a 3-month VIX, contango versus backwardation), the *implied correlation* across index members, and the *skew* (the richness of downside puts relative to upside calls). Each factor sharpens a piece of the flow map.

The term-structure slope is the cleanest gate: front-end *backwardation* — near-dated implied volatility above longer-dated — signals acute near-term stress and overrides a naive "positive gamma means chop" read, because it is the signature of the negative-gamma, forced-selling regime. The *level* sets the baseline (a high level widens the expected move that all the impact figures scale against). *Volatility-of-vol* gates the vanna rally: a falling VVIX *and* a falling volatility level together are the condition under which the dealer vanna flow of Section 4 becomes a systematic bid; a falling VVIX alone is an incomplete tailwind. *Implied correlation* discriminates *clean* from *choppy*: high or rising correlation means the index members move in lockstep, the index-level flow read is strong, and trends are clean; low correlation (dispersion) means rotation under the surface, the index churns, and the index-level flow read weakens. And *skew* gates the downside: a steep put skew means the negative spot-volatility correlation is active and downside vanna is sharp (cascade fuel), while — counter-intuitively — a *flat* skew combined with negative gamma is the most dangerous combination, a cascade with no advance warning in the options market. The practical protocol is to read the regime from the gate, then read *who is pushing* from the flow map, and to treat the combination as a prior on the day's character — chop, grind, expansion, or cascade — never as a signal to be traded mechanically.

The *calendar phase* of Section 7 enters as a modifier on top of the gate. An expiry week biases toward *pinning* (a calendar force that can override the gamma read at high-gamma strikes); the week after a monthly expiry biases toward a *vacuum* in which trends run because the pinning gamma has rolled off; month- and quarter-end bring the rebalancing flows of Section 7, which make the first move of the session unreliable; and a VIX-expiry or a triple-witching day layers additional dated flows onto the map. The calendar does not move the regime by itself, but it changes *which* flow is loudest within it.

8.4 Identification limits — the failure catalogue restated for the underlying

Everything above is a prior, and the discipline of the backtesting paper requires that its limits be stated as plainly as its content. Four failure modes apply to *every* flow in this paper, and restating them for the underlying is the honest close of the map.

Flows are estimated, not observed. We do not see the dealer's book, the volatility-target industry's AUM, or the CTA's positions; we infer them from the option chain, from a control law applied to public prices, and from a public trend signal. Every inference rests on assumptions — the dealer-positioning assumption behind the gamma sign is the most fragile, and it can *invert* the sign of the estimated flow while the real flow is unchanged. A map drawn from estimated flows inherits the estimation error of each.

Flows overlap and cannot be cleanly attributed. On any given day, several forced flows are trading at once, and a move in the index is their *net*. Attributing a specific move to "the vol-target deleveraging" or "the CTA

flip" is, in general, not identified from price data alone; the flows are collinear precisely when they matter most (in stress, they all sell together), which is exactly when attribution is least possible.

Flows are endogenous. The trigger for each mechanical flow is itself a function of price. Volatility rises because prices fell; the CTA flips because the trend broke; the dealer's delta drifts because the underlying moved. Flow responds to price and price responds to flow, so none of these is an exogenous shock, and a regression of returns on estimated flow is contaminated by simultaneity. The clean identification designs — flip events, exogenous expiry roll-offs, instrumented flows — are the subject of the dealer-flows-and-GEX paper and the overfitting-control paper, not of this map.

Flows are non-stationary. The composition of the flow landscape changes. The rise of 0DTE options since 2022 has reshaped the intraday gamma and charm profile; the AUM of the volatility-target and risk-parity complex has grown and shrunk over cycles; the crowdedness of trend-following waxes and wanes. Evidence from one epoch may not transfer, and a footprint measured in-sample may — as the turn-of-month test showed directly — decay out-of-sample. The map must be re-estimated, not assumed constant.

The correct posture toward the map is therefore the one the backtesting paper prescribes for any result built on estimated, overlapping, endogenous, non-stationary data: treat it as a *prior* that organizes attention and generates hypotheses, hold every reading to the direction check, and never let a plausible mechanism graduate to a validated signal without an out-of-sample test it could have failed.

9 From flow map to testable claims

A map that only organizes attention would be a poor return on a research-grade paper. Its real payoff is that each flow, once stated with a sign and a timing, becomes a *pre-registered, falsifiable hypothesis* — a one-sentence claim with a directional alternative, a pre-committed effect size, an out-of-sample split, and a decision rule fixed before the data can answer, in the exact format the backtesting paper requires. This section makes the hand-off explicit and describes the reproducible free-data pipeline that supports it.

9.1 One hypothesis per flow

Each flow family yields at least one confirmatory claim, and the author's own hypothesis registry already carries several of them in pre-registered form.

- **Dealer gamma (Section 4).** *H1: days on which the estimated aggregate dealer gamma is positive show lower next-period realized volatility than days on which it is negative, after controlling for the previous day's realized volatility and the volatility level; the effect is monotone in the gamma magnitude and strongest near expiry.* This pair — the damping claim (H001) and its amplifying mirror (H002) — is the sharpest in the set because its sign is dictated by the mechanism, and the direction check is the whole test: an effect of the right magnitude and the wrong sign refutes it. The estimation of the gamma variable, and the flip-event and roll-off designs that address endogeneity, are handed to the dealer-flows-and-GEX paper; the multiple-testing correction across the gamma, charm, and vanna variants is handed to the overfitting-control paper.
- **Dealer charm (Section 4).** *H1: the sign of the aggregate charm flow, computed from the open-interest distribution in the early afternoon, predicts the sign of the late-session return better than chance on*

expiry days, and the close pins closer to the high-gamma strike than a random-walk benchmark implies (H003, conditional on the gamma sign). The control is ordinary intraday seasonality and closing-auction imbalance.

- **Volatility-target (Section 5).** *H1: days on which the reconstructed volatility-target rule deleverages by more than a pre-set threshold are followed by higher realized volatility and negative return continuation over the next session, relative to matched non-deleveraging days.* The reconstruction of Section 5 supplies the flow variable on free data; the endogeneity caveat of Section 8 makes the matched-control design essential.
- **CTA trend (Section 6).** *H1: in a pre-set window around a persistent flip of the 3/6/12-month sign ensemble, the index shows return continuation (for confirmed flips) distinct from a matched non-flip window.* The proxy of Section 6 supplies the positioning variable; the effect-size floor must exceed the transaction-cost drag of a flipping strategy (the cost curve of the backtesting paper).
- **Calendar (Section 7).** *H1: the turn-of-month window earns a higher mean daily return than the rest of the month, out-of-sample.* This one has already been run in Section 7, pre-committed, and returned a *correctly-signed but non-significant* out-of-sample result — a worked example of the pipeline reporting a near-null honestly, and a caution against trading a decayed calendar edge.

9.2 Two worked pre-registrations

To make the hand-off concrete rather than gestural, here are two of the claims written in the frozen pre-registration format the backtesting paper requires — the format that removes the freedom to decide what counts as a win *after* seeing the data. The first is the volatility-target claim, because its flow is fully reconstructible on free data and so it can be run end to end without any licensed input:

```

# Pre-registration – vol-target deleveraging predicts follow-through

## Confirmatory hypothesis (exactly one)
- H1: Days on which the reconstructed 10%-vol-target rule deleverages by more
    than 5% of NAV are followed by HIGHER next-day realized range and
    NEGATIVE mean next-day return, vs. matched non-deleveraging days.
- H0: After matching on the volatility level, deleveraging days carry no
    information about next-day range or return.

## Expected effect size
- Point expectation: +15% next-day range, -10 bps next-day mean return.
- SEWA: effects too small to clear one round-trip cost are treated as null.

## Data & sample (fixed, free)
- S&P 500 daily closes (Yahoo chart API, cached); rule as in figures/fig_voltarget.py.
- Period: 1990-01-01 .. last cached close.

## In-sample / out-of-sample split (fixed dates)
- In-sample (design & matching): <= 2015-12-31.
- Out-of-sample (opened once): >= 2016-01-01.

## Decision rule
- Support H1 iff both effects hold IN-SAMPLE with the pre-set sign, the sign
    holds OUT-OF-SAMPLE, and a volatility-matched control removes the trivial
    "high-vol begets high-vol" confound (endogeneity caveat, Section 8.4).

## Multiple-testing correction
- Deleveraging-threshold and horizon variants tried -> correction REQUIRED (overfitting-control paper).

```

The second is the dealer-gamma claim. It cannot be fully run on free data — the gamma variable requires the estimation apparatus of the dealer-flows-and-GEX paper and licensed chain data — so its pre-registration is written to hand off cleanly to that paper, with the direction check as the load-bearing clause:

```

# Pre-registration – dealer gamma sign gates next-period realized vol

## Confirmatory hypothesis (exactly one)
- H1: Positive estimated aggregate dealer gamma predicts LOWER next-period
    realized volatility than negative gamma, after controlling for RV(t-1)
    and the vol level; the effect is monotone in |gamma| (H001/H002).
- H0: Given RV(t-1) and the vol level, the gamma sign adds no information.

## Direction check (hard gate, decides the study)
- Reject H0 only if the SIGN matches the mechanism: positive gamma -> LOWER RV.
  A significant effect of the WRONG sign REFUTES H1 (the 2026-07 sign-error
  lesson); it is never read as "an effect, direction TBD".

## Data (handed to the dealer-flows-and-GEX paper; not free)
- Gamma estimate from the option chain per that paper's method (licensed); underlying
  intraday from a standard feed. No licensed value is published; only the
  sign and the test statistic are reported.

## Split, decision rule, correction
- Fixed-date OOS split opened once; monotonicity across gamma buckets required;
  flip-event and expiry-roll-off designs address endogeneity; DSR/PBO per the overfitting-control paper.

```

The two templates show the division of labour the whole library is built on: the flows that free data can reconstruct (vol-target, CTA, calendar) can be tested here and in the backtesting paper directly, while the flows that require estimation of an unobserved book (the dealer channels) are pre-registered here but *executed* in the dealer-flows-and-GEX paper under the overfitting-control paper's overfitting controls. In both cases the discipline is identical — freeze the question, name the sign, open the out-of-sample set once.

9.3 The reproducible free-data pipeline

The pipeline that produced every figure in this paper is the template for testing every hypothesis above, and it is deliberately austere. Prices come from the public Yahoo Finance chart API (daily S&P 500 and the long-Treasury ETF), fetched once and *cached* to local CSV so the figures rebuild offline and identically; the volatility complex, where needed, comes from the free CBOE and FRED series used in the volatility-surface paper. Each figure applies *one* rule, fixed in advance and written into the script's docstring, with *no* search over thresholds, windows, or assets — because the search is the data-snooping the paper warns against, and running it would make the author guilty of the sin the map is meant to expose. Every number quoted in the text is printed by the script that draws the figure, so the prose cannot drift from the code. The scripts are `figures/fig_flow_taxonomy.py`, `figures/fig_sqrt_impact.py`, `figures/fig_voltarget.py`, `figures/fig_cta_proxy.py`, and `figures/fig_turn_of_month.py`; each is self-contained and depends only on NumPy, Matplotlib, and the cached data. Promoting any of these reconstructions from an illustration to a *result* requires the further discipline of the backtesting paper — a frozen pre-registration, an out-of-sample split opened once, a cost-sensitivity curve, and the multiple-testing correction of the overfitting-control paper — and none of the figures here claims to have cleared that bar. What they establish is the *shape and sign* of each flow, honestly and reproducibly, which is exactly what a map is for.

Evidence basis

Every substantive claim in this paper rests on one of three kinds of basis, distinguished here so a reader can tell at a glance which is which — **own reproducible computation** (recomputed from free data or a self-contained model, reproducible via the cited script), **literature** (peer-reviewed article or textbook, cited inline), and **practitioner consensus** (desk practice with no clean citation, treated as the weakest). Nothing here is called *validated* on external sources alone; the stronger word requires an own out-of-sample computation.

Claim / result	Basis			
The square-root law predicts about 7.0 bps of impact at 1% participation versus about 1.8 bps for a calibrated linear line (a factor of about four), the two crossing near 20%	Own	reproducible	computation	— figures/fig_sqrt_impact.py
A 10%-vol-target rule on the S&P 500 spends most of its time deleveraged (mean allocation 0.74), cutting 31.7% of NAV in a day in Feb 2018 and 68% over 35 days in Mar 2020	Own	reproducible	computation	— figures/fig_voltarget.py
The 3/6/12-month trend ensemble reads +1.00 (S&P) and −1.00 (TLT) on 2026-09-17, is net long equities on 74% of days, and has flipped 32 times	Own	reproducible	computation	— figures/fig_cta_proxy.py
The turn-of-month premium is +7.6 bps/day in-sample (t = 2.07) but decays to +4.5 bps/day and loses significance (t = 1.00) out-of-sample	Own	reproducible	computation	— figures/fig_turn_of_month.py; OOS split
Aggregate equity demand is inelastic, with a multiplier of roughly five — a dollar of price-insensitive buying moves aggregate value by about five	Literature	—	(Gabaix & Koijen 2021)	
Metaorder price impact grows roughly as the square root of size and is robust across markets, epochs, and asset classes	Literature	—	(Tóth et al. 2011)	
Positive aggregate dealer gamma is associated with intraday reversal, negative gamma with momentum	Literature	—	(Barbon & Buraschi 2021)	
Volatility targeting reshapes the drawdown and higher-moment profile of an equity allocation	Literature	—	(Harvey et al. 2018)	
The time-series-momentum premium from which the CTA positioning proxy is built	Literature	—	(Moskowitz, Ooi & Pedersen 2012)	
Dealer-flow sizing (e.g. about 3% of futures volume to re-hedge a 1% move) is an order-of-magnitude desk figure, not an observed quantity	Practitioner	consensus	— not independently verified	
The sensitive AUM of the vol-target/risk-parity (about \$300bn) and CTA (about \$150bn equity sleeve) complexes are desk estimates that public data cannot pin down	Practitioner	consensus	— not independently verified	
The four-regime map (Chop / Grind / Expansion / Cascade) and the volatility-factor gate are the author's own daily framework	Practitioner	consensus	— not independently verified	

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